Computer Organization and Architecture (CSL251)
LAB-3
2025-26 W

RISC-V Assembly Programming Assignment

Instructions:

1. Write separate RISC-V assembly programs for each question.
2. Use the RISC-V GNU toolchain (riscv64-unknown-elf-gcc) to assemble and link your code.
3. Run your program using the Spike RISC-V simulator.
4. Take clear screenshots showing:
    a. Successful compilation
    b. Program execution in Spike
    c. Final output register values
5. Upload the following for submission under a single zip file:
    a. All assembly source files (.s files)
    b. Screenshots of execution and output
6. Clearly name your files using the format:
   Q1_reg.s, Q2_cpy.s, Q3_minelement.s, Q4_evenodd.s.

Question 1: copy registers

Write a RISC-V assembly program to use s0 and s1 as argument register for addition function.

Expected Output

First parameter s0 = 10

Second parameter s1 = 20

Output = 30

Question 2: Copy one Array to another Array

Write a RISC-V assembly program that use Function call for Copy One Array to Another
Task

- Pass source array, destination array, and size
- Function copies elements

Question 3: Find Minimum Element

Write a RISC-V assembly program that use Function call to Find Minimum Element in Array
Task

- Pass array base and size
- Function returns minimum value

Question 4: Count Even-Odd elements

Write a RISC-V assembly program that use Function call to count even-odd elements in an array.

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